

UNIVERSAL GEOMETRIC TOPOLOGICAL OPEN MODULAR SUBSTRATE

UGTOMS FORGE 1.0

Parts, Interfaces, Tolerances, Assembly and Inspection Cartridge

UGTOMS FORGE 1.0 architecture

Parts, interfaces and tolerances become proof-carrying assembly proposals.



A nominal fit is not a product release; inspection and governance remain explicit.

Pure domain evaluation may propose evidence; only the retained authority chain may commit state.

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ATTRIBUTION AND EVIDENCE BOUNDARY

Prepared for Tom Klootwijk. Requester attribution is recorded as supplied and is not independently verified. This is a technical design artifact, not legal proof, scientific validation, regulatory approval or production safety certification.

Contents and Reading Rule

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READING RULE	
The UGTOMS source line supplies the reusable authority, typing, order, evidence and packing discipline. The specialist domain model in this cartridge is an engineering-derived extension. Where the source corpus is silent, this report says so instead of presenting general domain knowledge as source-derived.	

Executive Decision and Release Scope

FORMAL DECISION

GO for FORGE as the third cartridge. Product definitions, interfaces, tolerance intervals, assembly dependencies, inspection evidence and lineage become one typed substrate, while mature CAD/CAM/FEA/metrology systems remain external oracles for their validated scope.

FORGE is designed for proof-carrying assembly rather than a replacement CAD kernel. A part may look correct, fit nominally and still be the wrong revision, material, tolerance state or process condition. The cartridge makes those failure modes explicit before an authoritative assembly or release event.

UGTOMS FORGE 1.0 architecture

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Pure domain evaluation may propose evidence; only the retained authority chain may commit state.

Figure 1. FORGE maps design, shop and inspection paths into one authority chain.

Source Basis and Evidence Discipline

The cartridge inherits the following UGTOMS commitments:

- Typed definitions and operators live inside the substrate, with stable IDs, dependencies and content hashes [S1-S3].
- Pure evaluation is upstream of support, compatibility, guard, verified proposal, deterministic commit and lineage [S1, S2, S4].
- Coordinates, pictures, sounds, meshes and packed codes are not identity or authority by themselves [S4-S8].
- Uncertainty, exactness, capability and error margins are explicit and may only be strengthened by evidence [S1, S2, S8].
- Component-scoped identities prevent a specialist domain pack from masquerading as a universal replacement of every other release [S10].

ENGINEERING-DERIVED DOMAIN CONTENT

The part, datum, interface, tolerance, assembly, process and inspection definitions are engineering-derived. No external CAD standard, GD&T standard, material database, torque table, building code, machinery directive, safety standard or regulatory requirement is imported by this PDF.

ID	Registered source	Use in this cartridge	Pages
S1	UGTS__FOUNDATION__UNICODE_v5.pdf	Foundation 5.0 OperatorCells, exactness/capability, Unicode literal/glyph/semantic separation, hot codebooks and authority boundary.	pp. 1-21
S2	UGTS_KC_4_2_General_Operator_Order_Addendum(1).pdf	Typed OperatorSpec, domains/codomains, units, laws, effects, dependencies, order profiles, certificates and open extension contracts.	pp. 2-18
S3	UGTS_KC_3_6_Tom_Klootwijk.pdf	Definitions as first-class content-addressed data, resolvable dependencies and explicit operation order.	pp. 1-14
S4	Unified_Geometric_Topological_Substrate_GPU_Native_Addendum.pdf	Canonical support → compatibility → guard → event → transition → lineage sequence; packed-state and performance evidence discipline.	pp. 2-15
S5	UGTS_KC_2_0_Tom_Klootwijk_Expanded_Substrate.pdf	Pattern, field, topology, kinematics, bounded dynamics and hybrid-event architecture.	pp. 3-15
S6	UGTS_KC_Two_Hands_3_0_Interactive_Graphics_Runtime.pdf	Stable identity, proposal verification, deterministic commit, checkpoints, replay and downstream presentation.	pp. 3-16
S7	UGTS_KC_3_9_KC_Elizabeth_Vector_Game_Runtime.pdf	Versioned projects, deterministic game-world snapshots/hashes, compact records and self-contained delivery.	pp. 2-16
S8	UGTS_KC_4_0_Spatial_Evidence_Ledger(1).pdf	Observation versus authority, uncertainty, provenance, atomic patches, checkpoints and evidence-bearing lineage.	pp. 3-14
S9	UGTS_KC_3_6_2_SCLP_Tom_Klootwijk.pdf	Finite packing, log-polar charts, interval guards, winding and explicit topology/rotation distinctions.	pp. 1-15
S10	UGTS_Versioning_Charter_Phase_1_Foundation_Chronology(1).pdf	Component-scoped semantic identities and explicit release graphs instead of one misleading global number ladder.	pp. 1-6

Open Modular Cartridge ABI

The cartridge is a concrete instance of the shared domain object D_FORGE:

D = (T, O, R, U, I, Q, E, G, Delta, L, Cert, K, Pi)	
Field	Contract
T	Entity, value and state types
O	Typed operators and transformations
R	Relations, graphs, topology and converse pairs
U	Units, dimensions, coordinate and profile conventions
I	Invariants, conservation laws and constraints
Q	Dependency, precedence, phase and reduction order
E	Observations, evidence, uncertainty and provenance
G	Guards, thresholds and admission predicates
Delta	Proposed and committed state patches
L	Lineage, replay, history and migration
Cert	Exactness, residual, interval, scope and status certificates
K	Cold atlas, hot codebook and packed representation
Pi	Optional visual, audio, physical or device projection

A conforming cartridge contains a manifest, type registry, operator registry, relation atlas, unit/profile system, dependency DAG, guard library, transition definitions, claims ledger, examples and validation fixtures. External mature libraries may sit behind adapters when they are safer or more accurate than a fresh reimplementation.

General Operator Record and Order

```
OperatorSpec = (id, version, syntax, arity, domains, codomain, units, shape,
               laws, exactness, effects, capabilities, dependencies,
               order_contract, invariants, provenance, content_hash)
```

The record follows the General Operator and Order layer [S2]. A symbol or glyph is not allowed to smuggle in undeclared type, unit, order, inverse or authority semantics.

EXAMPLE RECORD

forge.mate constrains two typed InterfacePort endpoints and returns a MateProposal plus residual/free-degree-of-freedom certificate. It may commit only when part IDs/revisions, datum frames, tolerance/clearance, tool/process and safety gates all pass.

Normative phase schedule

Phase	Meaning
1-3	parse, normalize and resolve content-addressed references
4-6	type, shape and unit validation; dependency closure
7-9	construct candidates; apply declared composition and reduction order
10-12	evaluate exact or bounded results; issue capability and residual certificate

Phase	Meaning
13	support, compatibility and guard classification
14	verified proposal and deterministic conflict resolution
15	atomic commit, lineage, replay and optional projection

Formal FORGE State

q_FORGE = (P, M, F, I, T, A, Q, E, L, U)

P parts/revisions M material/process profiles
 F datums/frames/geometry I interfaces and mates
 T tolerances/uncertainty A assembly plan and current state
 Q inspection/quality E evidence/provenance
 L lineage/replay U residual/status

Object	Authority role	Boundary
PartSpec	Stable part/revision identity, units, interfaces, geometry refs and governance.	A mesh, thumbnail or barcode is not the part identity by itself.
InterfacePort	Typed coupling endpoint with frame, profile, ratings and allowed mates.	Visual proximity does not imply compatibility.
AssemblyStep	Dependencies, tools, proposed transforms/mates, guards and effects.	A step cannot write state directly.
InspectionRecord	Measured value, method, instrument, calibration, uncertainty and decision rule.	An uncalibrated number is not conformance evidence.
ReleaseState	Governance stage reached through accepted events and required certificates.	A software status is not regulatory certification by itself.

IDENTITY SPLIT

part identity != world coordinate != geometry file != revision != inspection status != legal ownership. A part may move, be remeasured or change release status without becoming a new logical part, while a revision change may create an incompatible identity despite similar appearance.

Parts, Revisions, Materials and Frames

```
PartSpec = (id, revision, unit_system, geometry_refs, material_profile, mass_properties?,
            datum_frames, interface_ports, tolerance_profile, allowed_processes,
            safety_class, provenance, content_hash)
```

Geometry capability classes

Class	Meaning	Allowed claim
G0 spelling/reference	External file/feature identifier only.	Can resolve the intended representation.
G1 finite mesh/polyline	Bounded sampled geometry.	Discrete queries with declared error only.
G2 analytic/parametric	Versioned exact or bounded functions.	Exactness follows the primitive/profile.
G3 B-rep/CAD oracle	External mature kernel result.	Certificate includes kernel/version/tolerance/status.
G4 measured geometry	Metrology/scan evidence with uncertainty.	Observed feature state under the capture profile.

Frame contract

```
Frame = (id, parent_frame, origin, basis_or_rotation, handedness, length_unit,
         orientation_convention, uncertainty, provenance, hash)
```

SE(3) transforms from the retained kinematics layer [S5] may be exact symbolic values or bounded numerical values. Datum identity and frame lineage remain explicit; a transformed coordinate never replaces the part or feature ID.

Materials and process profiles

EXTERNAL-SOURCE REQUIREMENT

Material properties, allowable loads, process limits, fastener values, fire/chemical ratings and regulatory classifications must come from named authoritative sources. FORGE only carries their typed references, versions, applicability, evidence and non-claims.

Interfaces, Mates and Kinematic Constraints

Interface family	Required semantics	Typical compatibility checks
Mechanical mate	port frames, mating geometry, remaining DOF, contact/clearance profile	part/revision, shape, size, orientation, access, tolerance
Fastener role	fastener/feature profiles, process and inspection requirements	thread/size/material/profile, tool, authorization, evidence
Electrical	pin/net/signal/rating/keying profile	direction, voltage/current/domain, pinout, insulation, revision
Fluid	medium, pressure, temperature, seal/port profile	material, diameter, pressure class, orientation, contamination policy
Custom modular port	stable type ID, domain/codomain, invariants and adapters	exact type/profile match plus declared coercions

Mate proposal

```
MateProposal = (part_a, port_a, part_b, port_b, target_transform, constraint_set,
               tolerance_certificate, tool_process_profile, evidence_refs, priority, hash)
```

Constraint semantics

- A fixed mate may remove six relative degrees of freedom under a declared rigid profile.
- A revolute or prismatic mate preserves a typed motion subspace and limit/guard profile.
- A contact or clearance relation may be exact, interval-bounded or oracle-derived; capability is explicit.
- Disassembly is not the automatic converse of assembly. It requires its own support, access, safety and process profile.

Tolerance and Uncertainty Calculus

```
LinearTolerance = (nominal, lower_deviation, upper_deviation, unit, profile)
interval(x) = [nominal + lower_deviation, nominal + upper_deviation]
```

Worst-case interval arithmetic and statistical tolerance analysis remain distinct operators. The first provides guaranteed bounds under the declared intervals; the second provides a model-dependent estimate and must carry distribution/independence assumptions.

FORGE interval fit certificate

A safe toy example: interval arithmetic checks a declared clearance before assembly commit.

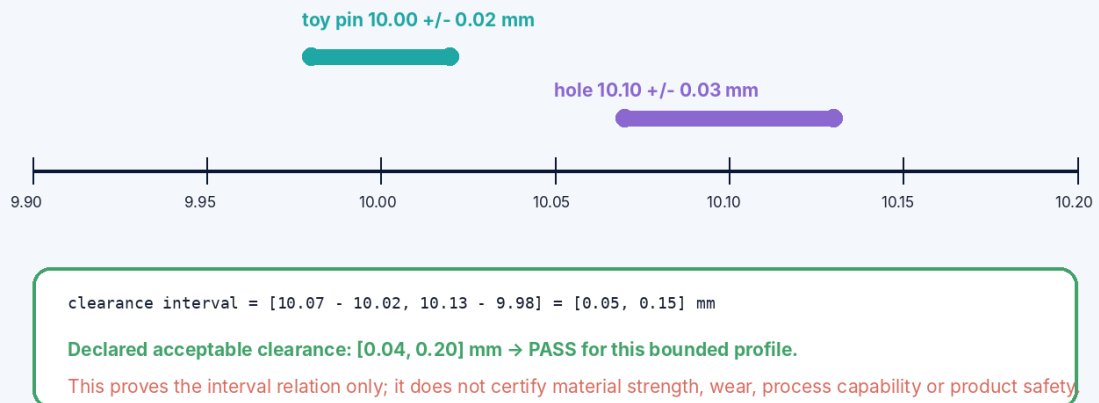


Figure 2. Exact interval arithmetic certifies a bounded toy clearance relation, not complete product safety.

Tolerance certificate

Field	Meaning
inputs	feature IDs, revisions, datums, nominal values, bounds and units
method	worst-case interval, statistical model or external CAD/metrology oracle
result	clearance/interference/position interval and residual
decision margin	distance between result interval and acceptance boundary
capability	exact under intervals, bounded numerical, statistical estimate or measured
provenance	source drawings, model/kernel/version, instrument/calibration and content hashes
STATISTICAL BOUNDARY	
A statistical stack may be useful for process planning, but it cannot silently replace a required worst-case safety check. The acceptance profile names which certificate class is admissible.	

Assembly Order and Atomic Commit

Assembly order is a dependency graph

Legal parts do not imply that every order is legal.



Attempting Final inspection before Pin aligned is rejected: dependency-unresolved.

Figure 3. The dependency DAG is part of the product definition; legal parts can still form an illegal sequence.

```

verified(step,q) = parts_present
                  and revisions_match
                  and interfaces_compatible
                  and dependencies_complete
                  and transform_residual <= margin
                  and tolerance_clearance_pass
                  and tool_process_ready
                  and safety_policy_pass
                  and conflict_free
  
```

Atomic assembly patch

Patch field	Meaning
added/removed mate relations	exact graph edits to assembly authority
part state changes	placed, aligned, fastened, inspected, quarantined, released
transform updates	new authoritative relative/world frames with certificate refs
consumed items	fasteners, adhesives, seals or other tracked inventory under profile
inspection obligations	new required measurements/checks generated by the step
pre/post hashes	canonical state identity before and after the commit

Conflict policy

- Two same-time steps touching the same mate/part state are ordered by declared priority and stable proposal ID.
- A rejected step records its first failed guard and does not partially mutate the assembly.
- A rework event adds a new versioned patch; it never deletes prior nonconformance or inspection history.
- Rollback restores a checkpoint only under an explicit governance policy and records the rollback event.

Inspection, Nonconformance and Evidence

Record	Required content	Authority boundary
Measurement	feature, value, unit, method, instrument, calibration, uncertainty, time, operator/profile	supports inspection; cannot be detached from calibration/profile
InspectionDecision	requirement ID/version, measured certificate, acceptance rule and reason	commits conforming/nonconforming state under declared scope
Nonconformance	affected part/feature, requirement, deviation, containment and evidence	preserves the original state and opens a corrective proposal
ReworkEvent	approved procedure/profile, affected state, evidence and post-inspection obligations	does not erase the nonconformance lineage
ReleaseEvent	all required certificates, approvals and open-obligation checks	software release state is still not a legal/regulatory certificate by itself
SAFETY-CRITICAL RULE		
For structural, medical, automotive, aerospace, pressure, lifting, electrical-safety, public-infrastructure or other safety-critical use, an implementation must bind authoritative standards, qualified professionals, material/process certificates and physical tests. Missing adapters block release.		

Repeat inspection and change

The Spatial Evidence Ledger discipline [S8] is reused: measurements are observations with uncertainty; accepted changes are deterministic patches tied to stable part/feature IDs. Repeat scans or inspections classify stable, changed, uncertain and unresolved features without treating a mesh difference as automatic truth.

Executable Engineering Notation

Literal / fallback	Typed meaning	Boundary
+/-	symmetric tolerance constructor	does not choose unit, confidence or statistical model
diameter / DIA	diameter feature modifier under a geometry/standard profile	glyph alone does not identify the datum or measurement method
parallel	parallelism relation between typed directions/features	visual parallel lines are not inspection evidence
perpendicular	orthogonality relation under a frame/profile	requires feature and datum context
position	profile-bound location tolerance relation	actual standard semantics come from an external versioned adapter
arrow / assembly step	ordered transition proposal	does not make the process safe or reversible
GLYPH AND STANDARD BOUNDARY		
Foundation 5.0 can co-address a symbol, canonical glyph and typed operator. FORGE does not copy or invent full engineering-standard semantics from the glyph. A standards adapter supplies the authoritative definition, edition, datum rules and conformance scope.		

Direct/converse relations

- part A mates to part B and part B receives part A may share one canonical relation after typed port normalization.
- assembly and disassembly are not generic converses; process, access and safety conditions differ.
- parent/child frame or contains/contained-by relations may use explicit converse surface spellings while preserving identity and lineage.

Deterministic Demonstration

```
toy pin: nominal 10.00 mm, tolerance +/-0.02 -> [9.98, 10.02]
hole:    nominal 10.10 mm, tolerance +/-0.03 -> [10.07, 10.13]
clearance worst-case: [10.07-10.02, 10.13-9.98] = [0.05, 0.15] mm
required clearance: [0.04, 0.20] mm -> PASS
```

Check	Result	Reason / certificate
pin/hole clearance interval	ACCEPT bounded mate	worst-case-clearance-pass
expected pin revision 3, actual revision 2	REJECT	revision-mismatch
attempt cap before pin alignment	REJECT	dependency-unresolved
measurement without calibration reference	REJECT inspection	calibration-missing
same geometry with different part ID	KEEP DISTINCT	identity-not-coordinate
change material profile without new evidence	REJECT release	certificate-scope-mismatch

Assembly queries

- `next_legal_steps(assembly_id)` → ordered proposals plus first-failure reasons
- `free_degrees_of_freedom(mate_graph)` → typed kinematic certificate
- `tolerance_path(feature_a, feature_b)` → dependency and interval trace
- `trace_part(part_id)` → revision, process, mate, inspection and rework lineage
- `open_obligations(release_state)` → missing inspections, certificates and approvals

Packing and adapters

Layer	Recommended treatment
Hot assembly record	part/interface local slots, transform/tolerance deltas, step state and integrity bits
Cold atlas	full PartSpec, revisions, geometry/material/process refs, requirements and hashes
CAD/CAM	external adapter returns geometry/process certificates and residual/status
FEA/physics	external adapter remains authoritative for declared load/model scope; results are evidence, not automatic release
Metrology	instrument/calibration adapters produce measurement records with uncertainty
Rendering/AR instructions	downstream projection; cannot bypass the verifier

Reference Validation, Boundaries and Kill Criteria

REFERENCE SELF-CHECK RESULT

24/24 deterministic specification self-checks passed. These checks cover catalog uniqueness, required fields, dependency closure, demo arithmetic and declared reason codes. They are package evidence, not independent domain validation.

Checked area	Result
Catalog integrity	48 unique FRG IDs; required fields present.
Interval arithmetic	Toy pin/hole fixture reproduces [0.05, 0.15] mm exactly.
Revision and dependency guards	Mismatch and out-of-order fixtures return deterministic reason codes.
Authority boundary	Geometry and proposed transforms cannot mutate assembly state directly.
Inspection evidence	Missing calibration/profile fixture rejects.
Adapter boundary	Safety-critical release requires external standards/qualification references.

Explicit non-claims

- No part, assembly, structure, machine, building, medical device or process is certified safe or manufacturable.
- No material property, fastener value, load case, torque, pressure rating or regulatory requirement is prescribed.
- No CAD/FEA/metrology standard conformance is claimed.
- No nominal or interval fit result proves fatigue life, wear, sealing, strength, reliability or process capability.
- No performance/compression advantage is claimed without implementation and equal-error measurement.

Kill criteria

1. Reject or narrow a use when a symbol, codebook slot or packed record cannot resolve to its exact global definition.
2. Reject when a capability or exactness label is stronger than the model, measurement or external source supports.
3. Reject when units, profiles, coordinate systems, operand order or dependency order are implicit.
4. Reject when a proposal bypasses support, compatibility, guards, deterministic commit or lineage.
5. Reject when an external adapter is required for safety or conformance but its version, calibration or provenance is absent.
6. Reject any performance, compression or safety claim without a named workload, target, error/equivalence contract and evidence.

Complete FORGE 1.0 Mechanism Catalog

The catalog contains 48 cartridge-local mechanisms. These IDs are component-scoped and do not silently extend a global M-number ladder.

ID	Family	Mechanism	Core contract
FRG001	Part	Part specification	Stable part ID, revision, units, interfaces, geometry references and provenance.
FRG002	Part	Revision identity	Revision is part of compatibility and cannot be inferred from appearance.
FRG003	Part	Material profile	Versioned material class/properties supplied by an external authoritative source.
FRG004	Part	Geometry reference	Parametric/mesh/B-rep reference with capability and tolerance status.
FRG005	Part	Mass-property record	Mass, center and inertia with method and uncertainty.
FRG006	Part	Safety class	Governance profile controlling required evidence and sign-off.
FRG007	Frame	Datum frame	Named coordinate frame tied to declared datum features.
FRG008	Frame	Local part frame	Stable local frame distinct from world position.
FRG009	Frame	Rigid transform	SE(3) transform with units, orientation convention and exactness.
FRG010	Geometry	Clearance relation	Minimum separation interval under declared geometry/tolerance profile.
FRG011	Geometry	Interference relation	Overlap/contact relation with capability status.
FRG012	Geometry	Reachability relation	Tool/part motion support under a bounded kinematic profile.
FRG013	Interface	Typed interface port	Mechanical, electrical, fluid or custom port with domain/codomain.
FRG014	Interface	Mate relation	Constrains relative frames and remaining degrees of freedom.
FRG015	Interface	Fastener relation	Fastener role and process requirements; values come from external sources.
FRG016	Interface	Thread profile	Standard/version-dependent thread semantics behind an adapter.
FRG017	Interface	Electrical interface	Pin/net/rating compatibility profile.
FRG018	Interface	Fluid interface	Port, medium, pressure and sealing compatibility profile.
FRG019	Tolerance	Linear interval	Nominal plus asymmetric or symmetric bounds.
FRG020	Tolerance	Angular interval	Angle/orientation bound with explicit convention.
FRG021	Tolerance	Geometric tolerance	Profile-bound feature-zone relation; glyph alone is insufficient.
FRG022	Tolerance	Worst-case stack	Interval propagation with no statistical assumptions.
FRG023	Tolerance	Statistical stack	Distribution/model-dependent estimate kept distinct from worst-case.
FRG024	Tolerance	Measurement uncertainty	Calibration, method and interval/coverage metadata.
FRG025	Guard	Part-present guard	Required stable part identities are available.
FRG026	Guard	Revision-match guard	Required and actual revisions are compatible.
FRG027	Guard	Interface compatibility	Port types, profiles and ratings may couple.
FRG028	Guard	Clearance/interference guard	Geometric relation and margin satisfy the assembly profile.
FRG029	Guard	Tool-readiness guard	Required tool, calibration and process profile are present.
FRG030	Guard	Safety-policy guard	External safety, regulatory and authorization requirements pass.
FRG031	Step	Place	Move a part into support without committing a mate.

ID	Family	Mechanism	Core contract
FRG032	Step	Align	Propose frame alignment and inspect residual.
FRG033	Step	Mate	Commit a declared constraint relation after verification.
FRG034	Step	Fastener apply	Process proposal requiring external values and evidence.
FRG035	Step	Inspect	Create a measurement/inspection proposal.
FRG036	Step	Release	Transition a verified assembly to the next governance state.
FRG037	Event	Assembly proposal	Candidate patch with required parts, dependencies and evidence.
FRG038	Event	Assembly commit	Atomic authoritative mate/state patch with pre/post hashes.
FRG039	Event	Disassembly	Reverse process only when an explicit supported procedure exists.
FRG040	Event	Measurement	Calibrated measurement appended to inspection lineage.
FRG041	Event	Nonconformance	Reason-coded divergence from drawing/profile/inspection requirements.
FRG042	Event	Rework	Versioned corrective proposal preserving prior lineage.
FRG043	Certificate	Tolerance certificate	Intervals, stack method, units, profile and decision margin.
FRG044	Certificate	Inspection certificate	Instrument, calibration, result, uncertainty and acceptance rule.
FRG045	Certificate	Process certificate	Qualified process/source/sign-off references; not generated from geometry alone.
FRG046	Packing	Part/interface codebook	Local aliases bound to cold atlas and revision hashes.
FRG047	Query	Assembly graph query	Dependencies, mates, free DOF, revision, inspection and lineage.
FRG048	Adapter	CAD/CAM/FEA/metrology adapter	Conventional mature tools remain authoritative for their declared scope.

Claims Ledger

Claim	Disposition	Reason
A nominal CAD fit proves manufacturability.	REJECT	Tolerance, process capability, material, surface, tool and inspection evidence remain required.
A mesh is the authoritative physical part.	REJECT	It is a representation with a declared capability and error contract.
Same coordinate means same datum or part identity.	REJECT	Stable IDs, frames, revisions and lineage remain distinct from position.
Worst-case and statistical tolerance stacks are interchangeable.	REJECT	They answer different questions and use different assumptions.
An uncalibrated measurement proves conformance.	REJECT	Instrument, calibration, uncertainty and method are part of the evidence.
A geometric-dimensioning glyph alone supplies complete standard semantics.	REJECT	Standard/version/profile and datum context must resolve explicitly.
Every legal assembly step commutes with every other.	REJECT	Dependencies, access and residual constraints make order result-affecting.
Rigid transforms can be exact within a symbolic profile.	ADMIT	Exact rationals/algebraic values or bounded floating profiles can be declared.
A content hash proves part quality, origin or legal ownership.	REJECT	It commits to bytes, not manufacturing truth or rights.
A seed reconstructs unstored material defects and wear.	REJECT	Physical evidence and uncertainty must be retained.
Software alone can issue structural or regulatory certification.	REJECT	Qualified external sources, professionals, standards and tests remain required.
This cartridge can coordinate mature CAD/FEA/metrology tools without replacing them.	ADMIT	Typed adapters preserve scope, version, inputs, outputs and certificates.

High-Impact Application Profiles

Profile	What the cartridge enables
Proof-carrying assembly manual	Every step exposes prerequisites, compatible interfaces, tolerance evidence, first-failure reason and replay.
Maker and educational construction kit	Typed sockets, order rails, tolerance cards and inspection tokens teach real engineering distinctions safely.
Digital thread and field repair	Stable part/revision identity connects work instructions, inspection, rework and offline evidence bundles.
Mass-customization configurator	Reject incompatible options before geometry generation and preserve the exact configuration lineage.
Multi-tool engineering orchestrator	CAD, CAM, FEA and metrology outputs pass through typed adapters and scoped certificates rather than one opaque file chain.

Final Formal Definition

FORMAL DEFINITION

UGTOMS FORGE 1.0 is a component-scoped fabrication and assembly cartridge in which part/revision identity, materials, datum frames, interfaces, tolerances, assembly dependencies, tool/process profiles, inspection evidence, nonconformance and release lineage are typed substrate data; every physical change is a verified atomic patch; and mature engineering tools, standards and qualified evidence remain authoritative for their declared scope.

Implementation sequence

7. Freeze JSON Schemas for PartSpec, InterfacePort, ToleranceCertificate, AssemblyStep and InspectionRecord.
8. Implement exact interval arithmetic, revision/dependency guards and atomic mate-graph commits as the scalar oracle.
9. Build a child-safe toy assembly demonstration with deterministic rejection and replay.
10. Add read-only CAD and metrology adapters with explicit kernel/version/calibration scope.
11. Require domain engineers, standards sources and physical validation before any safety-critical promotion.

RELEASE CONCLUSION

This 1.0 document freezes a domain-facing contract. It is deliberately useful before a production engine exists: implementations can now disagree visibly through typed profiles, reason codes and certificates instead of hiding differences behind the same symbol.